Feature Reconstruction Guided Fusion Network for Hyperspectral and LiDAR Classification

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***Abstract*— Deep learning has become increasingly popular in hyperspectral image (HSI) and light detection and ranging (LiDAR) data classification, thanks to its powerful feature learning and representation capabilities. However, HSI often contains substantial redundant information, which can hinder efficient data utilization. Furthermore, the significant disparity in information content between HSI and LiDAR data poses a major challenge in representing and aligning semantic information across these two modalities. To address these challenges, we propose a fusion network structure guided by feature reconstruction embedding (FRE). This approach employs feature decomposition to reconstruct HSI features and incorporates weight embedding to seamlessly integrate the reconstructed information into classification features. Furthermore, we introduce a cross-modal attention fusion module designed to merge extracted HSI and LiDAR features. This module fully exploits the complementary nature of these two type of feature, facilitating effective information exchange and semantic alignment across multimodal data. We evaluated our method on three widely used HSI and LiDAR datasets: Houston 2013, Augsburg, and MUUFL. Experimental results demonstrate that our proposed FRGFNet significantly outperforms traditional probabilistic methods and state-of-the-art deep learning networks, showcasing its effectiveness in multisource data fusion.**

***Index Terms*— Classification, fusion, hyperspectral images (HSIs), light detection and ranging (LiDAR), multimodel remote sensing.**

I. Introduction

In RECENT years, with the rapid development of remote sensing technology, remote sensing data has been increasingly used in various applications [1], [2]. Early research on remote sensing data classification relied mainly on a single data type, such as scene classification based on high-resolution remote sensing images and pixel classification using hyperspectral images (HSIs) [3], [4]. At present, HSI classification technology can produce satisfactory results. However, due to the characteristics of different objects with the same spectrum in HSI data, there are still certain limitations to using HSI data alone for land classification [5]. For example, in urban scenes, the pavements and roof materials of buildings may be similar, resulting in similar spectral curves. Therefore, using only spectral data to distinguish these land use and land cover types remains challenging [6].

Among various Earth observation methods, light detection and ranging (LiDAR) can generate digital surface model (DSM) products, making it easier to confirm the height of different objects and distinguish their categories. Due to the complementary nature of data characteristics with HSI, considering both HSI and LiDAR can further improve classification accuracy. In order to jointly utilize the rich discriminative information in HSI and LiDAR data corresponding to the same region, a large number of multimodal data fusion classification methods have been proposed.

Generally speaking, multimodel data fusion classification methods can be divided into three categories: pixel-level fusion, feature-level fusion, and decision-level fusion [7], [8], [9]. Zhao et al. [10] adopted the Gram–Schmidt method for preliminary pixel-level fusion of HSI and LiDAR. Liao et al. [11] proposed a feature fusion method to pair dimensionality reduction and the spectral information of the original HSI and morphological profiles. Hang et al. [12] proposed a decision-level fusion method that fully utilizes the ensemble characteristics of deep neural networks. Previous studies have shown that cross-modal interaction can improve cross-modal content understanding, while feature interaction can enhance the abundance of certain specialties. In the feature extraction process, a weighted method is used for feature interaction, while Mohla et al. [13] used LiDAR features as spatial attention to weight HSI features. Fang et al. [14] proposed a spatial–spectral enhancement module for cross-modal information interaction in deep neural networks. Li et al. [15] used a novel composite attention learning mechanism to fully integrate the features from both HSI and LiDAR data sources.

Deep learning methods have demonstrated powerful capabilities in feature mining and feature representation, making them widely applicable in various intelligent information extraction tasks for remote sensing images. Numerous studies

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have shown that deep learning techniques offer significant advantages over traditional methods, particularly in handling complex and high-dimensional data. In the field of image classification based on HSI and LiDAR, deep learning has achieved remarkable results [16]. Researchers have proposed various innovative network architectures to improve feature extraction and fusion.

For instance, Zhao et al. [17] introduced a fractional Fourier image transformer as a backbone network, which effectively extracts both global and local contextual information. Similarly, Song et al. [18] designed a network model that leverages the complementary nature of HSI and LiDAR data, optimizing the network structure and loss functions for enhanced performance. To overcome the limitations of naive networks, such as the lack of spatial feature construction capability and the locality of multimodal feature fusion, Li et al. [19] proposed a deep learning framework based on the MLP-mixer architecture. This approach improves the integration of multimodal data by enhancing spatial feature construction and fusion.

Additionally, Li et al. [20] employed diffusion-based posterior sampling to explicitly model joint information interactions between high-dimensional manifold structures in the spectral, spatial, and frequency domains. This method effectively eliminates the influence of masking texture effects in visual models, further refining the accuracy of feature representation. Recent advancements have also focused on multiscale and multisource fusion frameworks. Wang et al. [21] proposed a multiscale pyramid fusion framework based on a spatial–spectral cross-modal attention model. This framework excels in learning multiscale information, particularly in areas with complex information changes, thereby significantly improving classification accuracy. Similarly, Guo et al. [22] developed a multisource feature embedding and interaction fusion network specifically for coastal wetland classification, demonstrating the versatility of deep learning in handling diverse remote sensing tasks. In an innovative approach, Cao et al. [23] designed a spectral–spatial language fusion network that integrates visual and language features. By leveraging linguistic prior knowledge shared between spectral and spatial features, this network broadens the semantic space and enhances the interpretability of classification results. Building on this, Yang et al. [24] utilized class text information to guide the learning of visual representations. They established semantic associations between text and visual features through a multidimensionally incorporated contrastive learning paradigm, further advancing the field of multimodal classification.

Inspired by the core mechanism of HSI intrinsic image decomposition, Yu et al. [25] explored its potential for multimodal remote sensing data fusion. Their work delves into the inherent modality connections between HSI and LiDAR data from an intrinsic perspective, offering new insights into the fusion of heterogeneous data sources.

Due to the limitations of imaging mechanisms, HSI data often suffer from feature redundancy and low signal-to-noise ratio. Therefore, in HSI information processing, dimensionality reduction is often used to preserve more effective information [26]. Due to the lower information content of LiDAR data compared to HSI data, dimensionality reduction and fusion with LiDAR data have become a feasible approach. Liao et al. [11] implemented a graph-based fusion algorithm integrating dimension reduction techniques and geometry and spectral information fusion of the HSI and LiDAR data. Li et al. [15] considered the redundant information presented in the original HSI data, and principal component analysis (PCA) is used for spectral dimensionality reduction in our experiments. Guo et al. [27] combined graph networks and convolutional networks for classification, and performed dimensionality reduction on HSI data at the input of graph structures. Zhao et al. [28] used PCA to extract the principal components to reduce the spectral dimension. Wang et al. [29] believe that PCA will distort the spectral continuity of the HSI data, but 3-D convolution can still generate more discriminative feature maps from the spectral dimensions than 2-D convolution.

Many methods introduce the idea of dimensionality reduction into HSI and LiDAR classification to address the issue of feature redundancy in HSI data. Using dimensionality reduction methods to extract key information from HSI data in classification tasks is a direct approach. By selecting appropriate bands for HSI data, feature transformation can be achieved, which can more effectively enhance the representation ability of features. However, existing methods often consider these two parts separately: they first perform dimensionality enhancement or reduction on the data for effective feature reconstruction, followed by feature extraction and classification to categorize the data. With the increasing complexity of network modules in recent years, it has become technically feasible to combine feature reconstruction and feature extraction. Unfortunately, existing work has not yet considered these two parts as a whole.

Owing to HSI data are correlated, the assumption of relative independence between different bands is not satisfied because of HSI data. The features reconstructed by PCA are more expected to be distributed along the axis of the principal components, but such way does not fully reconstruct a feature space that is more conducive to classification.

In addition, when combining dimensionality reduction with deep learning methods, the complex mapping relationships inherent in deep learning often undermine the mathematical meaning of feature representation.

After dimensionality reduction, many methods use neural networks to process each reduced feature with equal weights but do not take into account the weights learned from the dimensionality reduction methods. For example, the meaning of each principal component in PCA is that each channel represents the information content of the original data. Introducing this information into the feature extraction and fusion framework can provide good guidance for feature extraction using a feature reconstruction model.

In this articles, we introduce an innovative approach to feature extraction and fusion through a reconstruction-guided fusion mechanism [30]. Unlike traditional dimensionality reduction techniques such as PCA or graph-based methods [11], [15], which rely on linear projections or neighborhood structures, the matrix decomposition provides a more flexible framework for capturing complex interactions

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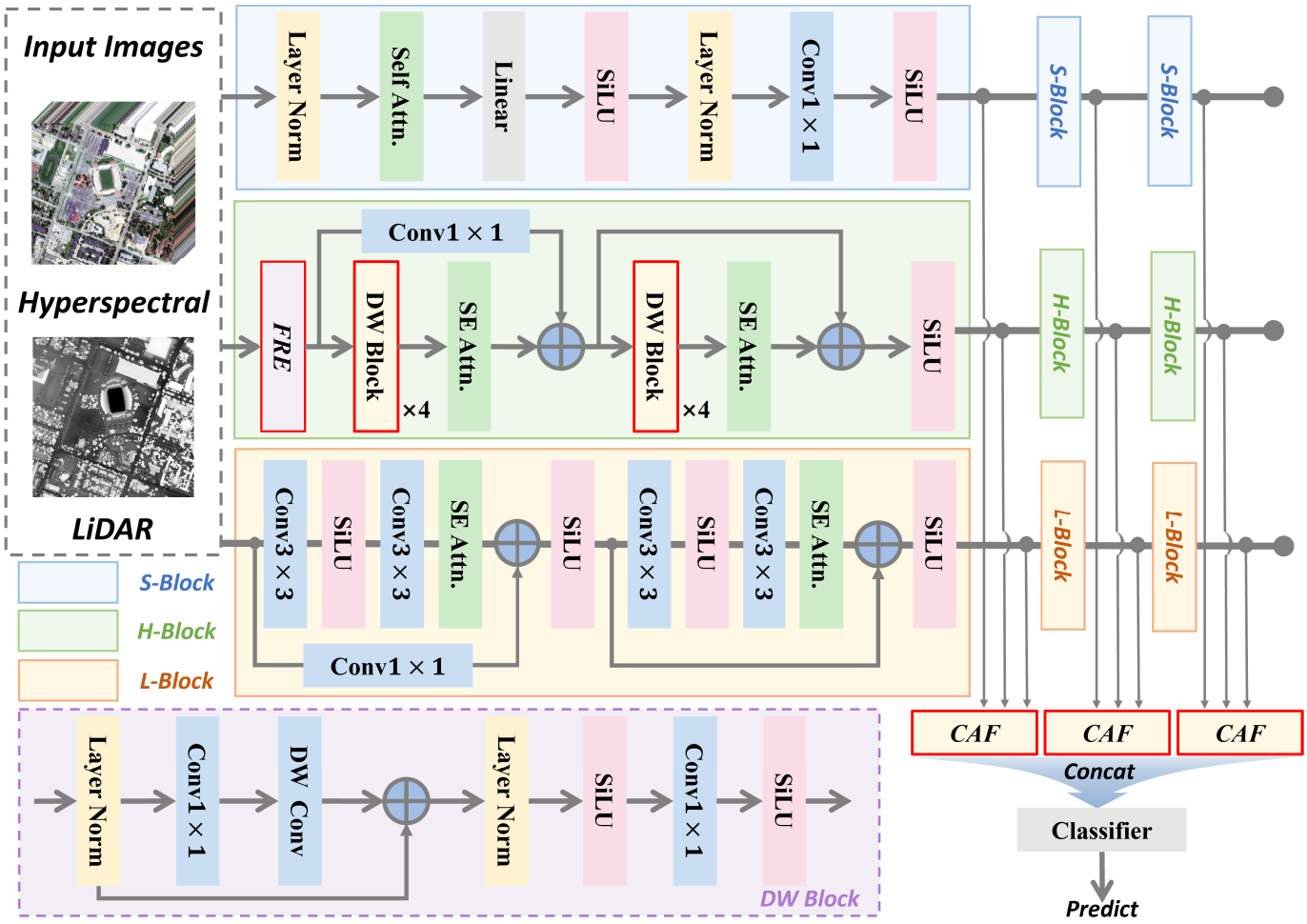


Fig. 1. Overall framework of the proposed FRGFNet for the HSI and LiDAR classification. *H*-block is for extract HSI spatial feature, *S*-block is for extract HSI spectral feature, and *L*-block is for extract LiDAR image feature.

between features. While PCA typically reconstructs features along the axes of principal components, often overlooking nonlinear relationships, matrix decomposition enables the extraction of more meaningful feature components by decomposing the data into several latent factors. This approach enhances the method's robustness and improves its ability to capture intricate patterns present in high-dimensional data.

For the classification task of combining HSI and LiDAR, how to effectively fuse the two parts of the features is currently the mainstream research focus. Although both types of data are essentially raster data, there are significant differences in the surface information they reflect, and the amount of information contained in the two types of data is also unequal. However, existing classification and fusion methods still have shortcomings in their thinking and design from this perspective. Compared to HSI data, LiDAR data only serve as a supplement, and it is difficult to classify land features using LiDAR data alone. Therefore, we hope to use HSI data as the backbone and cross-fuse LiDAR data at various levels of feature extraction.

The main contributions of this model are as follows.

1) We propose a novel HSI and LiDAR data classification framework called feature reconstruction guided fusion network (FRGFNet), which integrates feature reconstruction, feature extraction, and feature fusion as a whole. The scheme can effectively utilize reconstruction information to guide the classification process.
2) To obtain a more fitting reconstructed feature space for hyperspectral data, we propose a matrix decomposition-based feature reconstruction method to enhance the spectral features in hyperspectral data.
3) We propose a multiscale cross-modal attention fusion module, which utilizes LiDAR data to provide complementary features for hyperspectral data with a focus on specific aspects.

II. Proposed Method

In response to the challenges identified in HSI and LiDAR classification, our proposed approach, FRGFNet, illustrated in Fig. 1, consists of a feature reconstruction embedding (FRE) module, several feature extraction blocks, and a cross-modal attention fusion module.

A. Problem Formulation

In this article, the HSI is represented by $H \in \mathbb{R}^{C_0 \times H_0 \times W_0}$ and the corresponding LiDAR image is represented

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by $L \in \mathbb{R}^{H_0 \times W_0}$, where W_0 and H_0 represents the size of the two model images, respectively, and C_0 represents the spectral channels of HSI image. All pixels are assigned into $\mathcal{C}$ land cover classes, denoted by $\mathcal{C} = (y_1, y_2, \ldots, y_c)$. The classification method adopts the central pixel classification. Each patch is centered at pixel (i, j) and has size $\tau_0 \times \tau_0$. As the network progresses, we will use C to represent the number of channels.

It is important to note that the patch size, denoted by τ_0, in this article refers to both the initial patch size and the feature map size before being processed through the network layers. As the network deepens, τ_0 gradually reduces to τ, where $\tau \leqslant \tau_0$. When extracting spectral features, the feature map needs to be flattened spatially, defined as $\zeta_0 = \tau_0^2$. Similarly, as the network progresses, we will use ζ to represent the number of feature samples.

B. FRE Module

In the process of embedding the network using feature reconstruction methods, we adopted a new nonlinear decomposition method inspired by the Gaussian process latent variable model (GPLVM) called FRE module. This process includes two steps: estimate the correlation matrix K from the observed value X and identify the latent variable Z from the estimated correlation matrix K.

The first step can be solved by estimating K with the unbiased estimator, i.e., sample covariance $\rho = (1/(N-1))(X - \overline{X})(X - \overline{X})^{\mathrm{T}} \approx (1/(N-1))XX^{\mathrm{T}}$, where $\overline{X} = (1/N)\sum_{n=1}^{N} X_{:,n}$ ought to be 0 since $X_{:,n}$ are i.i.d. samples from a zero-mean Gaussian as

$$X_{:,n} \sim \mathcal{N}(0, K) \quad \forall n \in \{1, 2, \ldots, N\} \tag{1}$$

where K is a $T \times T$ covariance matrix generated by evaluating the kernel function k of the Gaussian process at all pairs of rows in Z, i.e., $k_{i,j} = k(z_i, z_j)$ where z_i and z_j are the ith and jth rows of Z.

Therefore, our main focus is to estimate the latent Z given ρ in the second step. We defined a squared exponential kernel as $k(z_i, z_j) = \eta^2 \exp(-(||z_i - z_j||^2/2l^2))$, where η^2 is the marginal variance and l is the length-scale. We defined $d_{i,j} = (||z_i - z_j||^2/l^2)$, so that $d_{i,j} = -2\ln(k_{i,j}/\eta^2)$. Writing $d_{i,j}$ in the matrix form $D = (d_{i,j})_{T\times T}$, we have

$$D = \begin{bmatrix} 0 & \cdots & (z_1 - z_T)^{\mathrm{T}}(z_1 - z_T) \\ \vdots & \ddots & \vdots \\ (z_T - z_1)^{\mathrm{T}}(z_T - z_1) & \cdots & 0 \end{bmatrix}. \tag{2}$$

To map D to the latent variable Z, we define a function θ with $\theta(D) = ZZ^{\mathrm{T}}$. By solving for d representing ZZ^{T}, we could write the matrix form $ZZ^{\mathrm{T}} = (z_i^{\mathrm{T}} z_j)_{T\times T}$ as

$$ZZ^{\mathrm{T}} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & d_{2,1} & \cdots & \frac{1}{2}(d_{2,1} + d_{1,T} - d_{2,T}) \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \frac{1}{2}(d_{T,1} + d_{1,2} - d_{T,2}) & \cdots & d_{T,1} \end{bmatrix}. \tag{3}$$

This is a symmetric semi-positive definite matrix with a unique matrix factorization. We will summarize the entire FRE process as follows:

$$\rho = \frac{1}{N-1}(X - \overline{X})(X - \overline{X})^{\mathrm{T}}. \tag{4}$$

which serves as an estimator of the covariance K

$$\eta^2 = \frac{1}{T}\sum_{i=1}^{T} \rho_{i,i}. \tag{5}$$

Estimates η^2 through a statistic of the diagonal of ρ

$$\hat{D} = -2\ln\frac{\rho}{\eta^2}. \tag{6}$$

Serves as an estimation of D. Finally, we can do an eigen decomposition on $\theta(\hat{D})$

$$\theta(\hat{D}) = U\Lambda U^{\mathrm{T}}. \tag{7}$$

To obtain the eigenvalue U and eigenvector Λ, which serve as estimations for the latent variable Z. Specifically, the decomposed feature vector Λ is multiplied by the original input matrix X to generate the reconstructed input, which is then fed into the model. Additionally, the decomposed eigenvalues U are concatenated with the reconstructed data to perform feature encoding, ensuring the retention of information features within the channels. That is, the output of the FRE module is $\mathcal{F} = \mathrm{Concat}(U, X\Lambda)$.

C. Feature Extraction Network

1) Hyperspectral Spatial Feature Extraction Block: In HSI classification, it is common to extract spatial and spectral features separately from HSIs. Unlike general natural image data, hyperspectral data exhibit strong correlations between adjacent channels. As previously mentioned, the FRE module reconstructs the information from the original data and reinforces the mapping relationships between data channels as much as possible. To maintain these mapping relationships, we utilize depth-wise convolution as the primary framework for feature extraction.

In H-block process, for an input feature $X_H \in \mathbb{R}^{C\times\tau\times\tau}$, if a complete feature extraction block is represented by the function f_H, then $y_H = f(X_H)$. Upon entering a specific feature extraction module, the input first passes through a designed gating mechanism. Next, feature extraction is conducted via depth-wise convolution, where adjusting the stride of this convolution effectively controls the dimensions of the input and output feature maps. After applying layer normalization and a sigmoid linear unit (SiLU) activation function, a point-wise convolution is employed to expand the features into additional channels, facilitating the diffusion of the network across the feature channels. The formula for this activation function is as follows:

$$\sigma(x) = \frac{x}{1 + \mathrm{e}^{-x}}. \tag{8}$$

To mitigate issues of gradient vanishing and exploding caused by stacking multiple layers, a residual connection is designed for the entire module. Following the residual module, layer normalization and a SiLU activation function are again

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applied to constrain the range of output features and enhance the nonlinear transformation mapping relationship. The overall workflow of this module can be formulated as

$$\phi_H(X_H) = \sigma(f_{\mathrm{dw}}(\mathcal{F}(X_H)) + f_{\mathrm{pw}}(\mathcal{F}(X_H)))$$
$$f_H(X_H) = \sigma(f_{\mathrm{dw}}(\phi_H(X_H)) + \phi_H(X_H)). \tag{9}$$

In this context, f_{dw} represents a feature extraction block consisting of four depth-wise convolution blocks arranged sequentially, along with a squeeze-and-excitation attention mechanism. f_{pw} denotes a point-wise convolution used to maintain a consistent number of channels, and σ represents the SiLU activation function. The SiLU activation function is not limited to this module; it is utilized throughout the entire network structure.

2) Hyperspectral Spectral Feature Extraction Block: Expanding the spectral dimension is an effective strategy in the process of extracting spectral features from HSIs. The data structure after unfolding is different from the usual visual task of stacking data on channels but rather represents the reflectance expressed by each pixel in different spectral reflectance intervals as the original feature input. For this unique structure, we adopt the self-attention structure as the main framework for feature extraction.

At the beginning of the S-block, to minimize the interference from pixels with long spatial distances on the central pixel, we crop the most central spectrum within the patch and unfold it. This results in each input feature $X_S \in \mathbb{R}^{\zeta \times C}$. The module designated for extracting spectral features from these input features is represented as f_S, where $y = f_S(X_S)$. First, we utilize a multihead attention mechanism to capture the long-range dependencies present in the features, expressed as

$$f_{\mathrm{mha}}(X_S) = \mathrm{Softmax}\left(\frac{QK^{\mathrm{T}}}{\sqrt{d}}\right)V. \tag{10}$$

Based on the aforementioned modules, we construct the spectral feature extraction structure proposed in this article. First, the input features undergo a layer normalization, followed by a multihead attention mechanism for feature extraction. A linear layer then maps each sample to a specified dimensional space, followed by a SiLU activation function. Subsequently, a combination of layer normalization, point-wise convolution, and SiLU is used to map the feature dimensions to the desired size. The entire process can be expressed as

$$f_S(X_S) = \sigma(f_{\mathrm{pw}}(\sigma(W \cdot f_{\mathrm{mha}}(X_S)))). \tag{11}$$

3) LiDAR Feature Extraction Block: The DSM product of LiDAR reflects only one feature channel on the digital image, so the information it contains is limited compared to HSI. A major challenge in feature extraction from LiDAR data is that increasing the number of channels to match that of HSI often leads to excessive feature redundancy. Consequently, it is essential to streamline the mapping relationships within the LiDAR data when designing the network.

Additionally, LiDAR data represent the DSM of the terrain, capturing elevation information. The extraction of elevation fluctuation features aligns more closely with traditional spatial feature processing methods used in images. Therefore, we utilize a conventional 3×3 convolution as the primary framework for feature extraction.

In L-block process, for an input feature $X_L \in \mathbb{R}^{C \times \tau \times \tau}$, we begin by employing a standard convolution module, followed by a layer normalization and a SiLU activation for feature extraction. In this step, the stride is constrained to control the size of the output feature. Next, we deepen the network structure using a configuration with a consistent stride of 1. Finally, a squeeze-and-excitation attention mechanism is applied to adjust the weight scores of each extracted channel. A residual structure is incorporated into this module to complete the LiDAR feature extraction framework. The entire process can be expressed as

$$\phi_L(X_L) = \sigma(f_{\mathrm{conv}}(X_L) + f_{\mathrm{pw}}(X_L))$$
$$f_L(X_L) = \sigma(f_{\mathrm{conv}}(\phi_L(X_L))) + \phi_L(X_L)). \tag{12}$$

The first extraction block, denoted as f_{conv}, consists of a convolution layer, a normalization layer, and a SiLU activation function. This is followed by a second extraction block, which connects another convolution layer, a normalization layer, and a squeeze-and-excitation attention mechanism.

D. CAF Module

Unlike the construction of classification models in a single modality that focuses on feature extraction and classifier design, in the classification task of multiple modalities, the sufficient and effective fusion of the features extracted by each modality is also a research focus. Specifically, in the classification task of combining hyperspectral and LiDAR, how to effectively fuse the spatial features extracted from HSIs, spectral features, and spatial features extracted from LiDAR can further improve the performance of the model. Due to the significant difference in the amount of information contained in hyperspectral and LiDAR, simply integrating them cannot effectively represent the potential relationship between the two modalities.

In order to address the issues raised above, this article proposes a novel fusion module called the cross-model attention fusion (CAF) module, which consists of multiple fusion modules that combine the data characteristics of each mode at the time of input. These modules can capture potential dependencies between different modes and effectively utilize the complementarity between features. The module architecture is shown in Fig. 2.

First, we set the input hyperspectral spatial features as $X_H \in \mathbb{R}^{C \times \tau \times \tau}$, spectral features as $X_S \in \mathbb{R}^{\zeta \times C}$, and LiDAR features as $X_L \in \mathbb{R}^{C \times \tau \times \tau}$. Besides, we used the nonlocal attention in this module. The process can be formulated as

$$\phi_{\mathrm{nl}}(x_1, x_2) = \mathrm{Softmax}(f_{\mathrm{pw}}(x_1) \times f_p(f_{\mathrm{pw}}(x_2)))$$
$$f_{\mathrm{nl}}(x_1, x_2, x_3) = \phi_{\mathrm{nl}}(x_1, x_2) \times f_p(f_{\mathrm{pw}}(x_3)) + x_3. \tag{13}$$

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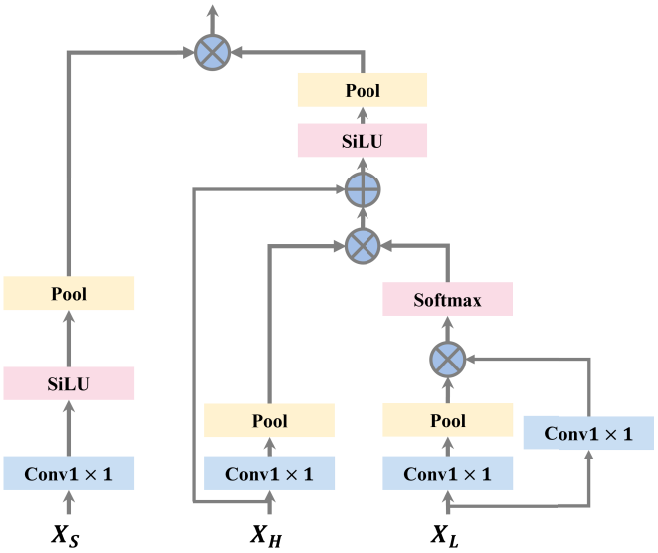


Fig. 2. Proposed cross-model feature fusion module framework.

Applied to our model, the overall module can be formulated as

$$f_{\text{caf}}^{i}\left(X_H^i, X_S^i, X_L^i\right) = f_p\left(\sigma\left(f_{\text{pw}}\left(X_S^i\right)\right)\right. \\ \left. \circ f_p\left(\sigma\left(f_{\text{nl}}\left(X_L^i, X_L^i, X_H^i\right)\right)\right)\right). \quad (14)$$

Among them, i represents the fusion performed at different scale levels. In this model, we use three modules of different scales to extract fused features of different scales from the image. f_p represents the adaptive average pooling layer.

Fused features obtained at different scales capture data characteristics at varying levels of detail. We fuse features across different scales through concatenation, leveraging each scale's unique characteristics for more comprehensive discrimination. The process can be formulated as

$$y_{\text{fe}} = \text{Concat}\left(f_{\text{caf}}^0, f_{\text{caf}}^1, f_{\text{caf}}^2\right). \quad (15)$$

E. Classifier and Loss Function

1) Classifier: The classifier maps the features extracted from the aforementioned modules to feature encodings corresponding to the number of categories $\mathcal{C}$, thus enabling the model to perform the classification task. The design of the classifier significantly impacts the overall performance of the model. Our classifier comprises three layers of perceptrons: each of the first two layers consists of a fully connected layer, a SiLU activation function, and a dropout layer, while the final layer contains only a fully connected layer.

2) Loss Function: In the multimodal learning of HSI and LiDAR, complex scenes typically exhibit a large number of categories but involve small areas, while extensive surface coverage often consists of a single category. This results in an imbalanced sample size within the dataset. To tackle this challenge, we adopted focal loss as the loss function

$$\mathcal{L}_{\text{FL}} = (1 - e^{-\mathcal{L}_{\text{CE}}})^{\gamma} \mathcal{L}_{\text{CE}}. \quad (16)$$

Among them, $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss. In this experiment, the value of γ is set to 2.

TABLE I
LAND-COVER CLASSES OF THE HOUSTON 2013 DATASET, WITH THE NUMBER OF TRAINING AND TESTING SAMPLES SHOWN FOR EACH CLASS

Class No.	Class Name	Training	Testing
1	Healthy Grass	198	1053
2	Stressed Grass	190	1064
3	Synthetic Grass	192	505
4	Trees	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking Lot 1	192	1041
13	Parking Lot 2	184	285
14	Tennis Court	181	247
15	Running Track	187	473
-	Total	2832	12197

III. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, we present three well-known HSI and LiDAR datasets: Houston 2013, Augsburg, and MUUFL. We begin by reviewing the current state-of-the-art methods and their implementation details as comparative approaches. Next, we conduct comprehensive experiments and ablation studies to quantitatively and qualitatively assess the performance of the proposed method for HSI and LiDAR image classification.

A. Dataset Description

1) Houston 2013 Dataset: In [31], this dataset was used for the 2013 IEEE GRSS Data Fusion Contest and collected using the ITRES CASI-1500 sensor over the campus of the University of Houston, Texas, and its surrounding rural areas. The image size is 349 × 1905 pixels, with a spatial resolution of 2.5 m. There are 144 bands in the wavelength range from 0.38 to 1.05 μm. This dataset contains a total of 15 029 ground-truth samples, and 15 distinguishable category labels are investigated in Table I. The visualization of the hyperspectral and LiDAR data in the Houston 2013 dataset and the distribution of the training and test samples in the spatial domain are shown in Fig. 3.

2) Augsburg Dataset: In [32], the Augsburg dataset was captured over the city of Augsburg, Germany. The hyperspectral data were acquired by the DAS-EOC HySpex sensor, and the LiDAR-based DSM data were collected by the DLR-3K system. The spatial resolutions of both the images were downsampled to a unified resolution of 30 m to adequately manage the multimodel fusion. In this dataset, the hyperspectral data consist of 180 bands ranging from 0.4 to 2.5 μm, while the DSM data have a single raster. The size of this dataset is 332 × 485 pixels. The dataset depicts seven different land-cover classes. In Fig. 4, the first image illustrates a false-color composite image of the hyperspectral data, followed by a gray-scale image of

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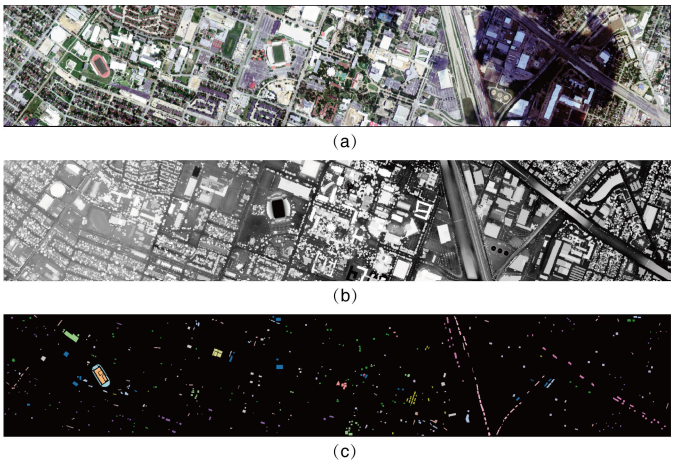


Fig. 3. Houston 2013 dataset. (a) False color composite illustration of HSI. (b) LiDAR-derived rasterized DSM. (c) Ground truth.

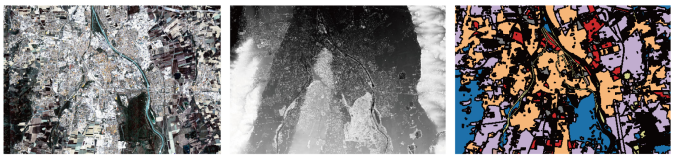


Fig. 4. Augsburg dataset. (a) False color composite illustration of HSI. (b) LiDAR-derived rasterized DSM. (c) Ground truth.

TABLE II

LAND-COVER CLASSES OF THE AUGSBURG DATASET, WITH THE NUMBER OF TRAINING AND TESTING SAMPLES SHOWN FOR EACH CLASS

Class No.	Class Name	Training	Testing
1	Forest	146	13361
2	Residential Area	264	30065
3	Industrial Area	21	3830
4	Low Plants	248	26609
5	Allotment	52	523
6	Commercial Area	7	1638
7	Water	23	1507
-	Total	761	77533

the LiDAR data, and finally, the ground-truth map is displayed as shown in Table II.

3) MUUFL Dataset: This dataset was taken at the University of Southern Mississippi Gulfport Campus, Long Beach, Mississippi, USA, in November 2010. The data collection contains hyperspectral and LiDAR-based DSM over the campus. The hyperspectral was collected by the ITRES Compact Airborne Spectrographic Imager (CASI-1500) sensor and LiDAR data were captured by the ALTM sensor using a laser at 1064 nm wavelength. Hyperspectral is composed of 325 × 220 pixels with 64 available spectral channels ranging from 0.38 to 1.05 μm. The spatial resolution of hyperspectral is 0.54 × 1.0 m, while that of LiDAR-based DSM is 0.60 × 0.78 m. This dataset contains a total of 53 687 ground-truth samples, and 11 distinguishable category labels. The distribution of randomly selected 50 samples from each class is shown in Table III. In addition, a visualization of the spatial distribution of the hyperspectral and LiDAR data

TABLE III

LAND-COVER CLASSES OF THE MUUFL DATASET, WITH THE NUMBER OF TRAINING AND TESTING SAMPLES SHOWN FOR EACH CLASS

Class No.	Class Name	Training	Testing
1	Trees	50	23196
2	Mostly Grass	50	4220
3	Mixed Ground Surface	50	6832
4	Dirt and Sand	50	1776
5	Road	50	6637
6	Water	50	416
7	Building Shadow	50	2183
8	Building	50	6190
9	Sidewalk	50	1335
10	Yellow Curb	50	43
11	Cloth Panels	50	129
-	Total	550	53137

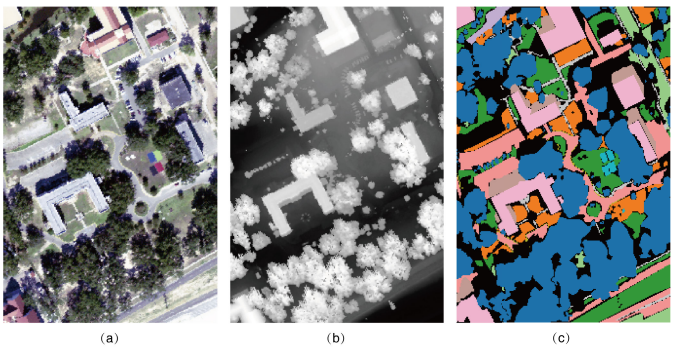


Fig. 5. MUUFL dataset. (a) False color composite illustration of HSI. (b) LiDAR-derived rasterized DSM. (c) Ground truth.

and the training and test samples in the MUUFL dataset is presented in Fig. 5.

B. Experimental Setup

1) Implementation Details: Our proposed FRGFNet is implemented using the PyTorch platform and optimized using the AdamW optimizer. For learning rate adjustment, a step strategy was employed, dynamically reducing it by half every 30 epochs. The initial value of the learning rate is 0.0001. The maximum number of training epochs for the networks was configured at 300. Batch normalization was integrated into the process, and a batch size of 16 was selected for the training phase. All experiments were carried out using a NVIDIA GeForce RTX 3060 graphics card.

2) Compared Methods: We selected several state-of-the-art baseline methods for comparison, including traditional machine learning methods such as KNN and SVM. Deep learning methods included HCT [28], DSHFNet [33], CALC [5], MsFE-IFN [22], M2FNet [34], and CrossHL [16].

3) Evaluation Indicators: We have used several widely used metrics, such as each class accuracy, overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ). These metrics provide a comprehensive evaluation of the classification performance of our model. Higher values for these metrics generally indicate better classification accuracy in remote sensing image classification tasks.

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TABLE IV

CLASSIFICATION ACCURACIES (%) AND KAPPA COEFFICIENTS OF DIFFERENT MODELS ON THE HOUSTON 2013 DATASET. THOSE BEST ACCURACIES ARE SHOWN WITH THE BOLD TYPE FACE

Class No.	KNN	SVM	HCT [28]	DSHFNet [33]	CALC [5]	MsFE-IFN [22]	M2FNet [34]	CrossHL [16]	FRGFNet
1	82.62	82.72	82.43	**83.10**	82.05	**83.10**	**83.10**	82.34	82.85
2	82.89	84.02	95.87	84.40	84.87	83.84	96.99	85.15	**100.00**
3	**100.00**	99.01	99.80	98.42	87.72	95.45	99.60	98.61	97.96
4	91.95	91.67	98.01	92.99	89.21	92.90	97.25	**99.24**	98.75
5	97.35	98.96	**100.00**	99.91	**100.00**	**100.00**	99.24	**100.00**	99.62
6	95.10	92.31	99.30	83.22	99.30	95.80	99.30	93.01	**100.00**
7	77.52	77.15	80.50	82.93	86.57	**94.87**	92.63	88.25	88.36
8	42.07	51.00	80.82	88.89	78.92	82.62	91.55	79.11	**96.53**
9	66.48	79.32	68.74	67.33	**92.16**	83.10	92.07	88.57	80.65
10	50.10	32.53	**92.76**	73.46	64.87	48.36	61.97	51.54	81.88
11	74.29	73.43	90.13	67.84	**99.81**	83.02	85.86	96.11	77.67
12	73.10	55.14	**99.81**	91.84	91.45	92.99	93.18	85.50	91.52
13	60.70	38.60	90.18	92.28	90.53	87.37	85.61	**92.98**	92.22
14	**100.00**	**100.00**	**100.00**	97.57	98.79	98.38	95.95	95.55	**100.00**
15	98.31	98.10	**100.00**	**100.00**	**100.00**	**100.00**	94.50	94.08	**100.00**
OA (%)	76.40	74.77	90.14	85.05	88.02	86.08	90.22	86.97	**90.89**
AA (%)	79.50	76.93	91.89	86.94	89.75	88.12	91.25	88.67	**92.54**
κ	0.7461	0.7280	0.8929	0.8379	0.8704	0.8492	0.8940	0.8585	**0.9011**

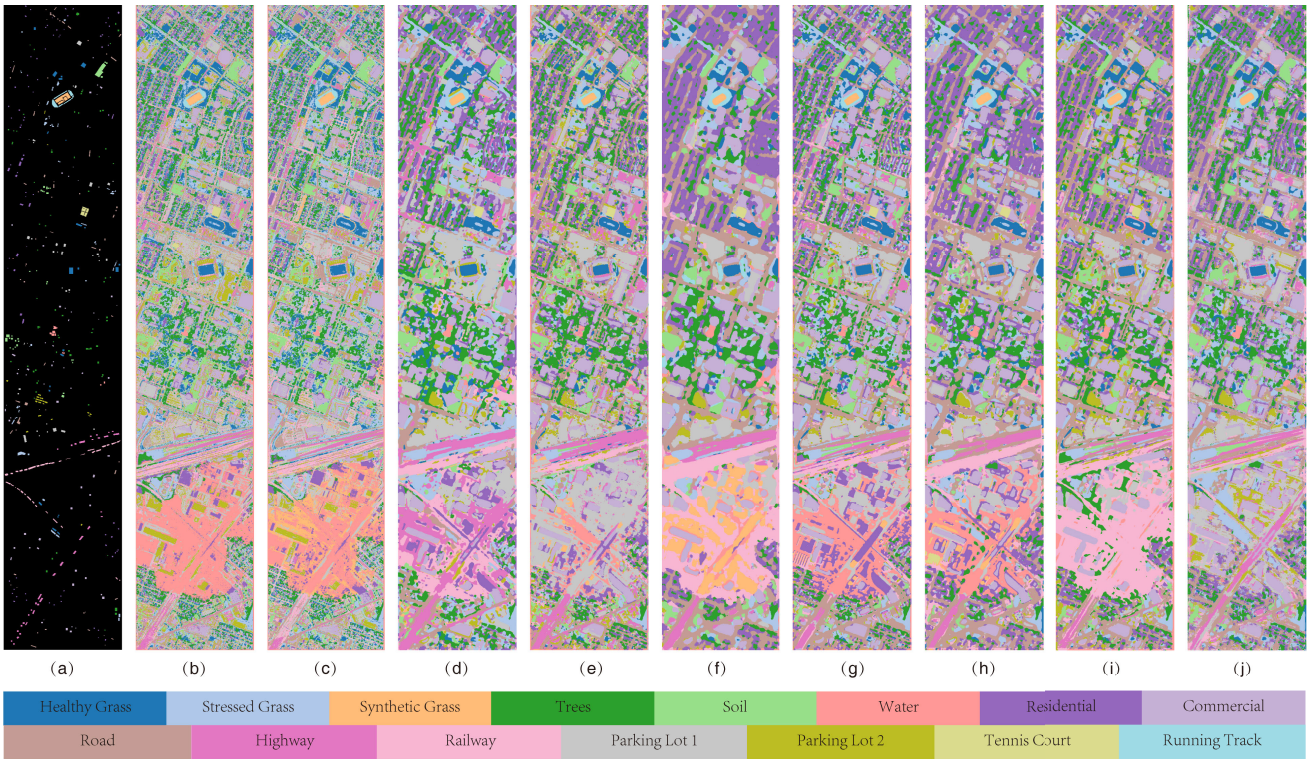


Fig. 6. Classification maps of different methods on the Houston 2013 dataset. (a) Ground truth. (b) KNN. (c) SVM. (d) HCT. (e) DSHFNet. (f) CALC. (g) MsFE-IFN. (h) M2FNet. (i) CrossHL. (j) FRGFNet.

C. Experimental Results on Houston 2013 Dataset

The Houston 2013 dataset consists of 2832 training pixels and 12 197 testing pixels, covering a wide range of land cover categories such as grass, trees, soil, water, and a total of 15 categories. Table IV provides a comparative performance analysis of various algorithms on the Houston 2013 dataset. The last column of Table IV highlights the performance of our proposed method, which achieves the highest overall classification accuracy, average category accuracy, and Kappa coefficient compared to other methods on the Houston 2013 dataset. In addition, Fig. 6 shows the experimental results of our proposed method and other comparative methods, providing a more intuitive view of some visualized results.

The overall classification accuracy of several baseline traditional classification methods based on machine-learning KNN and SVM is 76.40% and 74.77%, respectively, indicating that using only spectral features and LiDAR elevation

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TABLE V
CLASSIFICATION ACCURACIES (%) AND KAPPA COEFFICIENTS OF DIFFERENT MODELS ON THE AUGSBURG DATASET. THOSE BEST ACCURACIES ARE SHOWN WITH THE BOLD TYPE FACE

Class No.	KNN	SVM	HCT [28]	DSHFNet [33]	CALC [5]	MsFE-IFN [22]	M2FNet [34]	CrossHL [16]	FRGFNet
1	89.93	75.35	**97.64**	97.40	94.74	97.10	93.32	95.02	95.78
2	82.61	86.99	97.90	96.26	96.44	97.58	**98.51**	97.71	96.79
3	11.67	15.74	50.37	60.63	89.40	72.06	49.79	61.10	**92.09**
4	66.25	75.05	95.37	**97.16**	94.79	83.96	91.95	96.08	94.79
5	32.50	2.68	**79.16**	48.76	29.25	57.17	40.92	30.78	75.50
6	9.52	1.71	4.88	9.83	0.00	1.16	11.84	**13.31**	0.00
7	43.26	43.20	48.71	**69.61**	46.32	37.36	30.92	52.02	48.29
OA (%)	72.10	74.14	91.59	92.34	91.77	88.08	89.42	91.75	**92.58**
AA (%)	47.96	42.96	67.72	68.52	64.42	63.77	59.61	63.72	**71.89**
κ	0.5934	0.6126	0.8787	0.8901	0.8815	0.8279	0.8455	0.8803	**0.8936**

features for image classification has significant limitations. The spatial features of the two data play a very important role in accurate image recognition. And the six deep learning-based classification network structures we used achieved accuracies of 90.14%, 85.05%, 88.02%, 86.08%, 90.22%, and 86.97% on this dataset. In contrast, the method proposed in this article guides the encoding information of feature reconstruction and considers the information of each modality, achieving an OA of 90.89%, which is significantly better than other methods.

The superior performance of FRGFNet can be attributed to its reconstruction-guided fusion strategy, which effectively integrates feature information from both spectral and spatial domains. Unlike M2FNet's multiscale 3-D–2-D CNN, which focuses on multiscale feature fusion across different resolutions, FRGFNet explicitly reconstructs the feature space before fusion, ensuring that critical information from each modality is preserved and enhanced. This reconstruction step not only strengthens the representation of complex patterns in the data but also helps in better aligning and merging features from different sources, leading to a more robust and discriminative feature set for classification. In comparison, M2FNet's multiscale approach processes features at different scales through 3-D and 2-D convolutional layers, which are effective for capturing spatial relationships but may fail to fully exploit the global context or maintain precise feature alignment.

On the other hand, CrossHL's transformer utilizes self-attention mechanisms to model long-range dependencies between features. While this approach excels at capturing global context and learning complex feature interactions, it relies heavily on the transformer's ability to weight different features appropriately. Moreover, the matrix decomposition in FRGFNet allows for a more precise extraction of feature components, which complements the fusion process by providing clearer representations of latent factors. This further strengthens its ability to capture intricate patterns in high-dimensional data, something that both M2FNet's multiscale CNNs and CrossHL's transformer might miss due to their more generic fusion or attention mechanisms.

In the classification tasks of various categories, our proposed method also demonstrated better performance, achieving the highest accuracy in five categories: stressed grass, water, commercial, tennis court, and running track. Our proposed method has good classification performance for distinguishing between healthy grass, stressed grass, and synthetic grass in grass. This reflects that our method fully considers the height information provided by LiDAR data. For the tennis court category, KNN and SVM based on probabilistic statistical models using spectral information exhibit excellent performance, which is a disadvantage of other deep learning methods. Our method fully utilizes the spectral information of hyperspectral data, resulting in excellent performance in this category.

D. Experimental Results on Augsburg Dataset

The Augsburg dataset consists of 761 training pixels and 77 533 testing pixels, including seven categories such as forest, plants, allotment, and water as shown in Table II. The experimental framework is consistent with the previous text. Table V presents a comparative analysis of the classification accuracy of several algorithms on the Augsburg dataset. The performance of the proposed method on the Augsburg dataset is shown in the last entry of Table V. The last three rows of data in Table V indicate that the proposed FRGFNet network outperforms the comparison methods in terms of overall classification accuracy, average category accuracy, and Kappa coefficient on the Augsburg dataset. In addition, Fig. 7 shows the experimental results of the proposed method and different comparative methods.

The Augsburg dataset is different from the Houston dataset and the MUFFL dataset in that its spatial resolution is only 30 m, therefore, it tends to classify land use rather than land cover. The most challenging categories in this dataset include industrial area, allotment, and commercial area, which are not only difficult to distinguish in terms of spectra. And the sample size provided is very small. Due to the lack of advantages in spectral information in land use classification systems, the classification accuracy of KNN and SVM only reached 72.10% and 74.14%, respectively.

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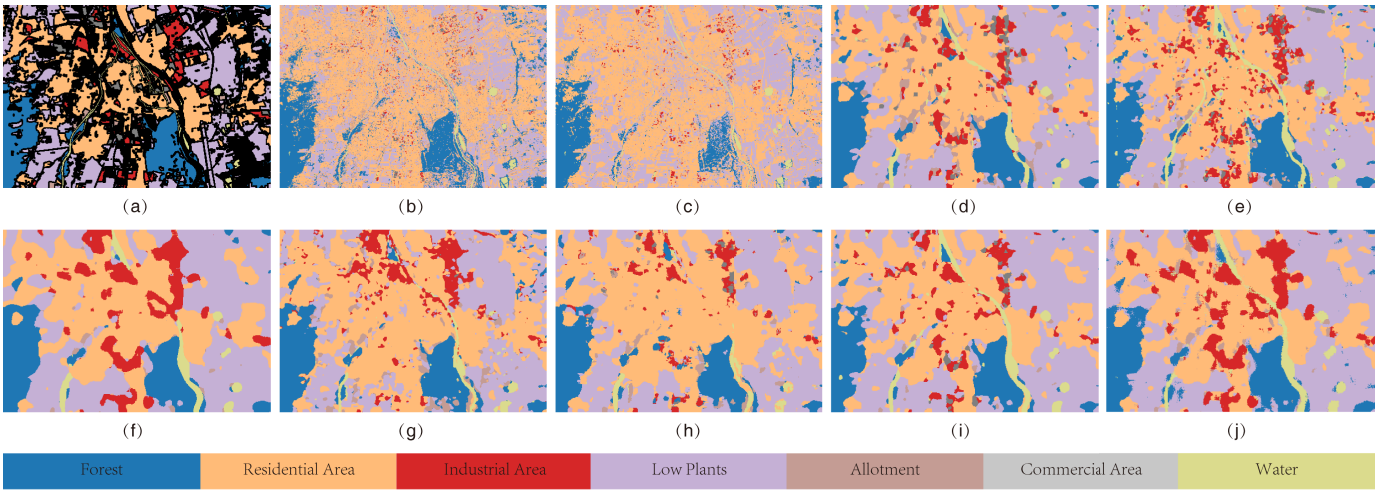


Fig. 7. Classification maps of different methods on the Augsburg dataset. (a) Ground truth. (b) KNN. (c) SVM. (d) HCT. (e) DSHFNet. (f) CALC. (g) MsFE-IFN. (h) M2FNet. (i) CrossHL. (j) FRGFNet.

TABLE VI

CLASSIFICATION ACCURACIES (%) AND KAPPA COEFFICIENTS OF DIFFERENT MODELS ON THE MUUFL DATASET. THOSE BEST ACCURACIES ARE SHOWN WITH THE BOLD TYPE FACE

Class No.	KNN	SVM	HCT [28]	DSHFNet [33]	CALC [5]	MsFE-IFN [22]	M2FNet [34]	CrossHL [16]	FRGFNet
1	89.50	77.80	92.26	91.76	91.67	93.99	91.26	**95.13**	94.04
2	**86.45**	81.40	80.66	79.36	70.21	80.88	69.17	74.69	85.98
3	53.29	58.27	65.53	69.35	72.53	74.05	70.43	79.45	**79.53**
4	77.93	73.93	92.29	**98.03**	97.92	95.44	94.43	95.78	94.99
5	86.36	86.18	84.38	83.22	72.94	81.08	91.59	89.47	**91.82**
6	94.47	91.59	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
7	75.90	36.37	82.87	90.66	91.80	90.88	91.34	94.41	**94.47**
8	84.98	87.53	94.59	92.76	**97.45**	95.90	95.25	94.93	95.63
9	42.47	57.45	59.18	31.99	62.47	74.68	52.51	74.46	**82.80**
10	60.15	68.42	75.94	86.47	**92.48**	90.23	36.09	87.22	84.96
11	93.15	94.52	**99.54**	**99.54**	**99.54**	**99.54**	97.71	98.63	98.82
OA (%)	81.54	75.57	86.02	85.69	85.42	88.49	86.43	90.27	**91.26**
AA (%)	76.79	73.95	84.29	83.92	86.27	88.79	80.89	89.47	**91.19**
κ	0.7595	0.6926	0.8167	0.8127	0.8107	0.8496	0.8229	0.8719	**0.8846**

Correspondingly, the accuracy of the six deep learning methods reached 91.59%, 92.34%, 91.77%, 88.08%, 89.42%, and 91.75%, respectively. Our method also has overall advantages in low spatial resolution hyperspectral remote sensing and LiDAR data.

In the Augsburg data, the main challenge lies in the issue of class imbalance, which is reflected in the four categories of industrial area, allotment, commercial area, and water. Especially for the commercial area category, which only has seven training samples, most methods are unable to achieve effective classification on this category, which has become a limitation of this dataset. For relatively small datasets, our classification method showed the best classification accuracy in the industrial area, and also demonstrated acceptable classification accuracy in the allotment and water categories, which is a stability that other methods do not possess. Similarly, due to the limitations of category labeling and scene selection in this dataset, even though our AA achieved the best performance, it still only had 71.89%, indicating that there is still a lot of room for improvement in the design of methods for this dataset.

E. Experimental Results on MUUFL Dataset

The MUUFL dataset includes 11 densely distributed land cover categories, some of which require more samples, posing challenges for their identification. Table VI presents a comparative analysis of the performance of various algorithms on the MUUFL dataset.

In this dataset, unlike the Augsburg dataset mentioned above, we uniformly sampled the MUUFL dataset and obtained an equal number of training samples. Note that traditional machine learning classification performs poorly in categories such as mixed ground surface, building shadow, sidewalk, and yellow herb. For example, building shadow does not essentially belong to a typical type of land cover, so it does not have obvious spectral features of land cover, which differs from our common sense of understanding of land cover. However, our classification method achieved a classification

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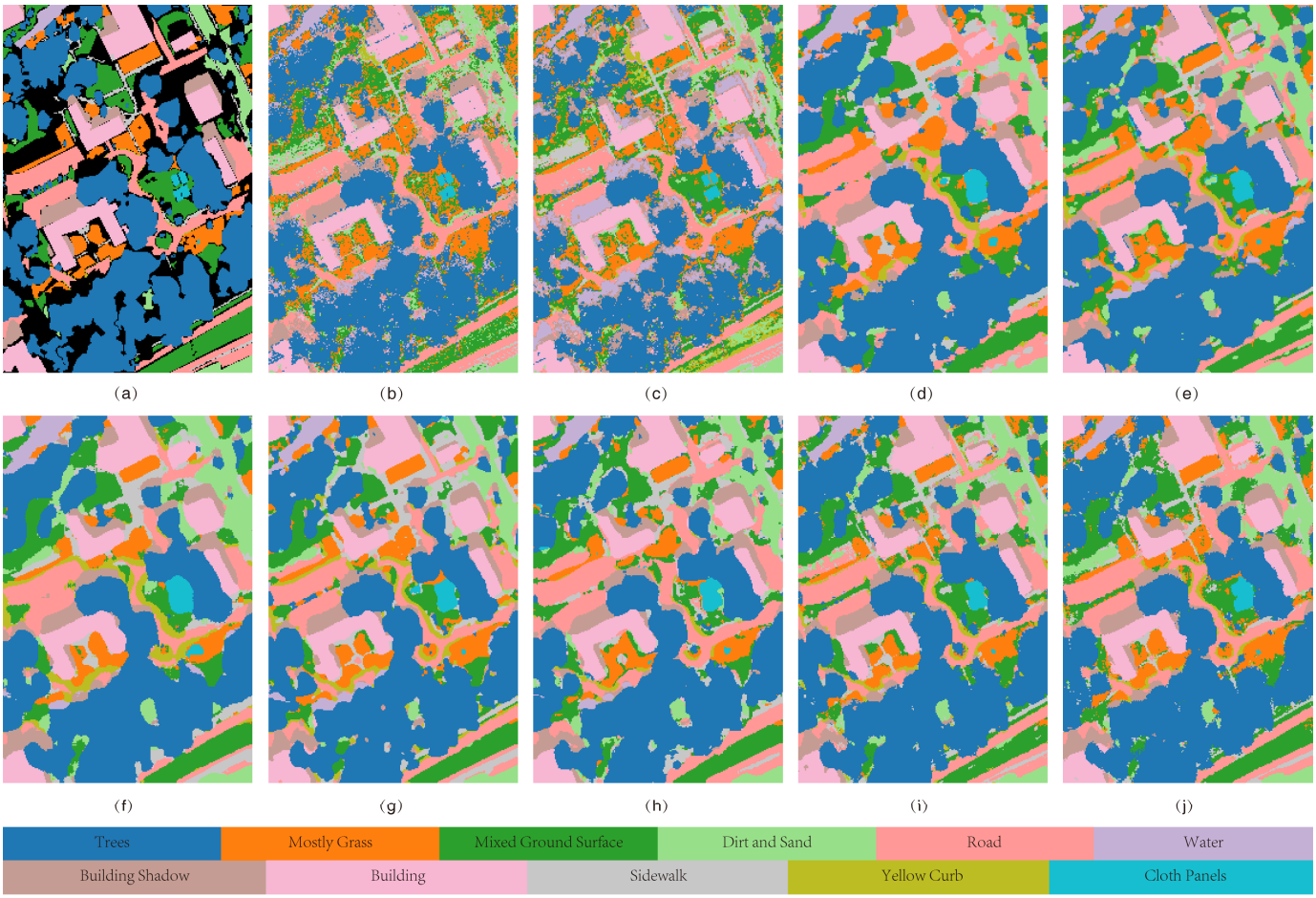


Fig. 8. Classification maps of different methods on the MUUFL dataset. (a) Ground truth. (b) KNN. (c) SVM. (d) HCT. (e) DSHFNet. (f) CALC. (g) MsFE-IFN. (h) M2FNet. (i) CrossHL. (j) FRGFNet.

accuracy of 94.47% in this category, which means that our model can break through this information barrier and achieve more refined classification. And our model also demonstrated excellent accuracy in several other easily confused categories. In addition, our model is also more accurate in outlining the road boundaries in the diagram. The classification map is shown in Fig. 8.

F. Model Analysis

We carry out various ablation experiments to study the impact of different configurations on the performance of the proposed FRGFNet.

1) Effectiveness of Different Image Patch Size: To assess the efficacy of the input modules, we conducted a systematic grid test to evaluate the input sizes for both the spatial and spectral streams. The experimental results are presented in Fig. 9. The findings indicate that optimizing the input sizes for each stream to match their respective network structures leads to the best performance. This tailored approach to input sizes significantly enhances overall network performance.

2) Ablation Study: To validate the necessity of the proposed method, we conducted an ablation study on different modules within the model, as illustrated in Table VII.

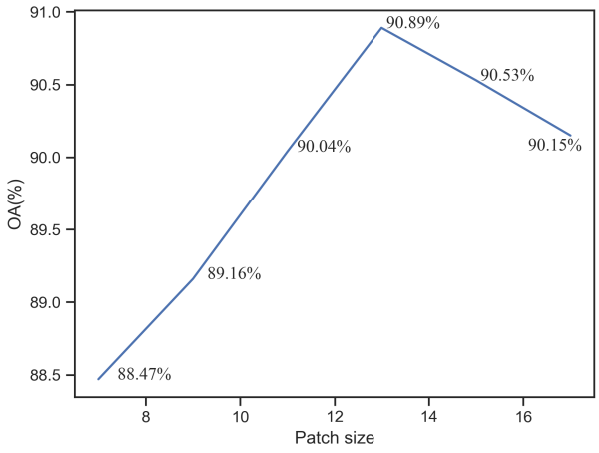


Fig. 9. Classification results achieved by different patch size for the Houston 2013 dataset.

For a fair comparison, we ensured that the hyperparameter settings remained consistent across different network configurations and employed three evaluation metrics to analyze the classification performance across three datasets. The ablation study focused on the two main components of FRGFNet: the feature reconstruction module and the feature

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TABLE VII
ABLATION COMPARISON OF EACH VARIANT IN FRGFNET

Module		Metrics	Datasets		
FRE	CAF		Houston 2013	Augsburg	MUUFL
✗	✗	OA(%)	90.01	91.79	88.53
		AA(%)	91.10	64.56	88.85
		κ	0.8913	0.8806	0.8509
✗	✓	OA(%)	90.24	92.30	90.22
		AA(%)	91.28	68.39	89.41
		κ	0.8945	0.8884	0.8699
✓	✗	OA(%)	90.35	92.33	90.41
		AA(%)	91.40	68.85	90.01
		κ	0.8953	0.8917	0.8733
✓	✓	OA(%)	90.89	92.58	91.26
		AA(%)	92.54	71.89	91.19
		κ	0.9011	0.8936	0.8846

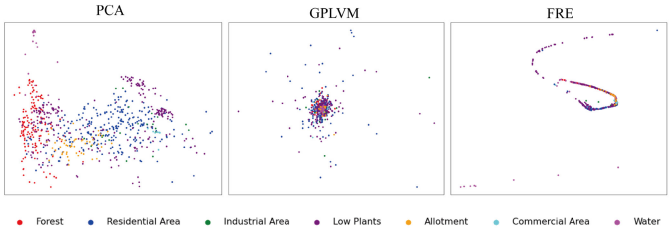


Fig. 10. Dimensionality reduction results achieved by different methods for the Augsburg dataset.

fusion module. The results indicated that considering only the feature reconstruction module improved OA by 1.32%, 1.24%, and 1.09% for the Houston 2013, Augsburg, and MUUFL datasets, respectively, compared to not including it. After integrating the fusion module, we observed improvements of 1.97%, 2.13%, and 1.26% across the three joint hyperspectral and LiDAR classification datasets. This enhancement is primarily due to our fusion module effectively facilitating communication between the two data modalities. The ablation study has demonstrated the effectiveness of each proposed module, significantly enhancing classification performance across the three evaluation metrics and enabling FRGFNet to better explore and differentiate between categories in practical scenarios.

3) Effectiveness of FRE: To further validate the effectiveness of the FRE module in reducing HSI redundancy, we conducted an additional comparative analysis. Specifically, we visualized the reconstructed features of the Augsburg dataset reduced to two dimensions using our FRE module and compared the results with those obtained from traditional dimensionality reduction methods, namely PCA and GPLVM as shown in Fig. 10. The visualization demonstrated that the FRE module not only effectively reduces redundancy but also preserves essential information more robustly compared to PCA and GPLVM. This comparison underscores the superiority of the FRE module in maintaining critical feature representations while minimizing redundant data, further validating its role in enhancing the overall performance of the proposed model.

TABLE VIII
CLASSIFICATION ACCURACIES (%) AND KAPPA COEFFICIENTS OF DIFFERENT FUSION METHODS ON THE HOUSTON 2013 DATASET. THOSE BEST ACCURACIES ARE SHOWN WITH THE BOLD TYPE FACE

Class No.	Concat	Add	CAF
OA (%)	90.35	90.03	**90.89**
AA (%)	91.40	91.12	**92.54**
κ	0.8953	0.8918	**0.9011**

4) Effectiveness of CAF: To further evaluate the effectiveness of the proposed CAF module, we conducted additional experiments comparing it with two simpler fusion methods: concatenation and element-wise addition. These experiments were performed on the Houston 2013 dataset to ensure a fair and focused comparison as shown in Table VIII. The results demonstrated that the CAF attention mechanism outperforms both concatenation and element-wise addition in terms of classification accuracy, highlighting its ability to more effectively integrate HSI and LiDAR data.

IV. CONCLUSION

In this article, we propose FRGFNet, an innovative approach for the joint classification of HSI and LiDAR data. FRGFNet employs feature decomposition to reconstruct HSI features and incorporates weight embedding to seamlessly integrate the reconstructed information into classification features. Extensive experiments on three benchmark datasets demonstrate that FRGFNet delivers superior classification performance, affirming the effectiveness of the proposed framework. Furthermore, ablation studies underscore the importance of the feature reconstruction module and the pivotal role of the CAF in boosting overall performance. Looking ahead, future research could focus on integrating pretrained foundation models to enhance cross-modal representation learning and improve semantic understanding. Moreover, leveraging self-supervised pretraining approaches could refine feature reconstruction while simultaneously reducing reliance on annotated data. These advancements would further strengthen the framework's capacity to model the intricate relationships between HSI and LiDAR modalities.

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